



A college under Mapúa Malayan Colleges Laguna

Exploratory Data Analysis of Machine Learning Solution Project

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I. Introduction

<Introduce the dataset and the purpose of the Exploratory Data Analysis. Write this section by answering the following guide questions:

- *What is the dataset about? Provide a brief overview.* The dataset contains information related to customer demographics, transactions, and social media interactions at FinMark Corporation. It aims to provide insights into customer behaviors, market trends, and interaction patterns across multiple platforms.
- *Why is this Exploratory Data Analysis important?* EDA is crucial for understanding the structure of the dataset, identifying missing or inconsistent data, and detecting patterns or trends that can help improve customer segmentation and business strategies.
- *What are the goals or objectives?* The goal is to gain insights into customer demographics, transaction patterns, and social media interactions and use these insights to enhance FinMark's marketing and customer segmentation strategies.
- *Who is the intended audience for this report?>* The intended audience includes business analysts, data scientists, and decision-makers at FinMark Corporation who are responsible for improving market trends understanding and customer segmentation strategies.

II. Dataset Overview

<Describe the basic characteristics and structure of the dataset. Write this section by answering the following guide questions:

- *How many variables (columns) and observations (rows) does the dataset contain?*
 - 'customer_demographics_contaminated.csv':
 - Number of Columns: 6
 - Number of Rows: 3200
 - 'customer_transactions_contaminated.csv':
 - Number of Columns: 6
 - Number of Rows: 3200
 - 'social_media_interactions_contaminated.csv':
 - Number of Columns: 6
 - Number of Rows: 3200
- *What are each variable's data types (numeric, categorical, etc.)?*
 - 'customer_demographics_contaminated.csv'
 - **Customer ID:** object/**Categorical**
 - **Age:** object/**Numeric**

- **Gender:** object/Categorical
- **Location:** object/Categorical
- **Income Level:** object/Categorical
- **Signup Date:** object/Categorical
- 'customer_transactions_contaminated.csv'
 - **CustomerID:** object/Categorical
 - **Transaction ID:** object/Categorical
 - **Transaction Date:** object/Categorical
 - **Amount:** object/Numeric
 - **Product Category:** object/Categorical
 - **Payment Method:** object/Categorical
- 'social_media_interactions_contaminated.csv'
 - **CustomerID:** object/Categorical
 - **Interaction ID:** object/Categorical
 - **Interaction Date:** object/Categorical
 - **Platform:** object/Categorical
 - **Interaction Type:** object/Categorical
 - **Sentiment:** object/Categorical
- *Are there any initial observations or peculiarities about the dataset's structure?>*

'customer_demographics_contaminated.csv'	● Customer ID: This is a unique identifier (UUID format), but there were duplicates found. ● Age: I found unrealistic values like -1 and 150; while there were also some marked as 'Unknown' and 291 null values. ● Gender: All were consistently tagged as either 'Male' or 'Female' ● Location: A town name found to be a possible misspelling: e.g. Adamshire vs Adamsshire; Castillofort vs Castilloport ● Income Level: Consistent naming and formatting (Low, Medium, High) although there were a 303 null values ● Signup Date: Consistent dd/mm/yyyy formatting; range is from July 2019 to June 2024
'customer_transactions_contaminated.csv'	● Transaction ID: This is a unique identifier. There should be no duplicate transaction IDs but I found a total of 200 duplicate records. ● Transaction Date: Consistent dd/mm/yyyy formatting; range is from July 2022 to June 2024. ● Amount: There were unrealistic amounts like -100 and 0; while some were labeled as 'Free'. There were 304 nulls.

	- Product Category: Consistent naming and formatting (Automotive, Clothing, Electronics, Health & Beauty, and Home & Garden); while some were left blank (299). - Payment Method: Consistent naming and formatting (Bank Transfer, Credit Card, Debit Card, PayPal); no blanks.
'social_media_interactions_contaminated.csv'	- CustomerID: Consistent text formatting - Interaction ID: This is a unique identifier. There should be no duplicate transaction IDs but I found a total of 200 duplicates. - Interaction Date: Consistent dd/mm/yyyy formatting; range is from July 2023 to June 2024 - Platform: Consistent naming and formatting (Facebook, Instagram, Twitter); but there were also some that were left blank (311). - Interaction Type: Consistent naming and formatting (Comment, Share, Like), no blanks. - Sentiment: Consistent naming and formatting (Negative, Neutral, Positive, Very Negative, Very Positive, None) though with 329 missing values.

Customer ID, Transaction ID, and Interaction ID all have the same alphanumeric format; even the same length of characters.

Observations before first phase of deduplication	Observations after first phase of deduplication
- Customer Demographics has 177 duplicate rows. - Customer Transactions has 185 duplicate rows. - Social Media Interactions has 180 duplicate rows.	- Customer Demographics has 23 duplicate rows based on 'CustomerID'. - Customer Transactions has 15 duplicate rows based on 'TransactionID'. - Social Media Interactions has 20 duplicate rows based on 'InteractionID'.

Upon manual inspection of records that have duplicate unique IDs:

Observations	Decision
CustomerID duplicates (46 records) - either duplicate records have the same Age but different SignupDate (24 records) OR different Age but same SignupDate (22 records)	- For duplicate records that have the same Age but different SignupDate (24 records), I retained the one with the more recent SignupDate as it is more likely up-to-date. - For duplicate records that have different

	Age but same SignupDate (22 records), I picked the more reasonable age and deleted the records with either -1, Unknown, or 150 value.
TransactionID duplicates (30 records) - either duplicate records have the same TransactionDate but different Amount (16 records), different TransactionDate but same Amount (8 records), and varying TransactionDate and Amount (6 records)	1. For duplicate records that have the same TransactionDate but different Amount (16 records): a. Positive Amount vs "Free" - retained record with positive Amount since there was a PaymentMethod indicated b. Positive Amount vs Zero - retained record with positive Amount since there was a PaymentMethod indicated c. Positive Amount vs -100 - deducted the -100 against positive Amount since this could possibly be a voucher, discount, or a refund; provided that the positive Amount is bigger than 100. 2. For duplicate records that have different TransactionDate but same Amount (10 records), I retained the record with the more recent TransactionDate. 3. For duplicate records that have varying TransactionDate and Amount (4 records), I deleted both records since it was hard to determine which record is more reliable.
InteractionID duplicates (40 records) - either duplicate records have the same InteractionDate but different Sentiment (18 records) or different InteractionDate but same Sentiment (22 records).	1. For duplicate records that have different InteractionDate but same Sentiment (26 records), I retained the records with the more recent InteractionDate. 2. For duplicate records that have the same InteractionDate but different Sentiment (14 records), I decided to delete them both since there's no way to determine which record is more reliable.

III. Data Quality Check

<Assess and summarize the quality of the data. Write this section by answering the following guide questions:

- Which variables have missing values? How many missing values are there?

- 'customer_demographics_contaminated.csv'
 - **Age:** 291
 - **Income Level:** 303
- 'customer_transactions_contaminated.csv'
 - **Amount:** 304
 - **Product Category:** 299
- 'social_media_interactions_contaminated.csv'
 - **Platform:** 311
 - **Sentiment:** 329

- What methods were used to handle missing data (e.g., imputation, removal)?

First, I identified records that have more than 1 missing data and then removed them. For those records that only have one missing data, I performed mean imputation for numeric variables while mode imputation was applied to categorical variables.

- What are the basic statistics (mean, median, min, max) for numeric variables?

Age		Amount	
count	2964.000000	count	2971.000000
mean	45.354926	mean	486.488112
std	18.582928	std	287.542303
min	-1.000000	min	-100.000000
25%	32.000000	25%	244.000000
50%	45.000000	50%	486.490000
75%	57.000000	75%	723.690000
max	150.000000	max	999.860000

- What are the frequency counts or distributions for categorical variables?

customer_demo graphics	Gender	Location	IncomeLevel	SignupDate
count	2964	2964	2964	2964
unique	2	2664	3	1460
top	Female	Port Michael	High	2020-11-09
frequency	1498	5	1183	8

<i>customer_transactions</i>	<i>CustomerID</i>	<i>TransactionDate</i>	<i>ProductCategory</i>	<i>PaymentMethod</i>
<i>count</i>	2971	2971	2971	2971
<i>unique</i>	1857	716	5	4
<i>top</i>	e54d1219-b6b3-4b5c-8b75-62f4148c23b6	2024-01-06	Clothing	Debit Card
<i>frequency</i>	6	11	822	755

<i>social_media_interactions</i>	<i>CustomerID</i>	<i>InteractionDate</i>	<i>Platform</i>	<i>InteractionType</i>	<i>Sentiment</i>
<i>count</i>	2966	2966	2966	2966	2966
<i>unique</i>	1884	366	3	3	5
<i>top</i>	4824bfa4-0e82-461c-a6ab-4b79b8b3ee39	2023-07-02	Instagram	Comment	Positive
<i>frequency</i>	5	16	1178	1001	1177

- *Did you identify any outliers? How were outliers handled or interpreted?*

Outliers	Frequency	Interpretations	Handling
A negative value for age (-1) customer_demographics	26	clearly an error, as age cannot be negative; values are likely to be data entry errors (e.g., a typo or miscapture)	Deletion - Since negative age represents a small fraction (0.88%) of the dataset
Extremely high age (150) customer_demographics	31	clearly an error, as oldest person verified was aged 122; values are likely to be data entry errors	Deletion - Since negative age represents a small fraction (1.05%) of the dataset

		(e.g., a typo or miscapture)	
A negative value for amount (-100) customer_transactions	33	seems to represent an invalid transaction (as negative amounts are typically not valid unless there was a refund or a discount).	Review if customers have other transactions that are of the same or close transaction date and of the same category. Deletion - Since negative age represents a small fraction (1.11%) of the dataset, some customers don't have another transaction or while some show to have another transaction, the transaction dates are too far apart and/or the product category are different

IV. References

<Upload the Python code for the Exploratory Data Analysis to your Github Repository. Copy the share link and insert it in the table below.>

Reference No.	Reference
1	https://github.com/atomicbum170348/MO-IT125-S3103-Cabalin/blob/main/Milestone1_EDA/socmed_behavior.csv https://github.com/atomicbum170348/MO-IT125-S3103-Cabalin/blob/main/Milestone1_EDA/transaction_behavior.csv
2	Python Code for Exploratory Data Analysis https://github.com/atomicbum170348/MO-IT125-S3103-Cabalin/blob/main/Milestone1_EDA/Milestone%201%20EDA%20v1.ipynb

